

Stony Brook University ILL

ILLiad TN: 951436



Borrower: VVH

Lending String: NYG,NYG,VVU,MU9,*VZB

Patron:

Journal Title: The Journal of hand surgery.

Volume: 12 **Issue:** 6

Month/Year: 1987-11**Pages:** 1041-3

Article Author: American Society for Surgery of
the Hand Richman, J A; Gelberman, R H; Engber,
W D; Salamon, P B; Be

Article Title: Ganglions of the wrist and digits:
results of treatment by aspiration and cyst wall
puncture.

Imprint: New York, NY : Elsevier

ILL Number: 235293953



Call #: The Journal of hand surgery.

Location: jstks

ODYSSEY ENABLED

Charge

Maxcost: 0.00IFM

Shipping Address:

Daemen University Library ILL
4380 Main St
Amherst, New York 14226
United States

Fax:

Ariel: 198.22.176.100

Email: library@daemen.edu

Ganglions of the wrist and digits: Results of treatment by aspiration and cyst wall puncture

In a prospective study, 87 carpal and digital ganglions were aspirated, multiply punctured, and digitally ruptured. Fifty percent of wrists and digits were immobilized for 3 weeks and 50% were mobilized early. Mean follow-up was 22 months. Thirty-six percent (31/87) of all ganglions treated showed a successful outcome. Twenty-seven percent (16/60) of dorsal carpal, 43% (6/14) of palmar carpal, and 69% (9/13) of palmar digital ganglions did not recur. Immobilization significantly improved the results of treatment of dorsal carpal ganglions. Forty percent (12/30) of those in the immobilization group and 13% (4/30) of those in the early mobilization group had a successful outcome ($p < 0.05$). (J HAND SURG 1987;12A:1041-3.)

Jonathan A. Richman, M.D., Richard H. Gelberman, M.D., William D. Engber, M.D., Peter B. Salamon, M.D., and Delois J. Bean, M.D.,
San Diego and Stockton, Calif., and Madison, Wis.

A number of conservative measures for the treatment of carpal and digital ganglions have been advocated, including manual rupture, cyst puncture with a needle, and the injection of steroids and sclerosing agents.¹⁻⁵ However, there have been no previous prospective studies that identify the patients who are the most likely to respond favorably to a single treatment method. Aspiration and cyst puncture, a technique useful in establishing the diagnosis, has been recommended as a method of definitive treatment. This study determines the results of the technique of aspiration-puncture, with and without immobilization, in a significant number of patients with wrist and digital ganglions.

Materials and methods

Between 1979 and 1984, 85 patients with 87 carpal and digital ganglions were evaluated prospectively: 29

at the University of California, San Diego Medical Center, and the Veterans Administration Hospital in San Diego, 22 at the University of Wisconsin Medical Center in Madison, and 34 at the private clinic of one of the senior authors (P. S.). For inclusion in the study all ganglions had to yield characteristic fluid on aspiration and completely decompress after multiple puncture. Patients with intra-articular pathology or rheumatoid arthritis were excluded. Prospective data sheets recording age, sex, duration of symptoms, time to recurrence, and follow-up were completed for each patient and sent to one of us for collation (R. H. G.). Success was defined as complete resolution of the mass. The mean duration of symptoms was 15 months (range 1 to 120 months). The average follow-up period was 22 months (range 6 to 46 months). Aspiration was performed in an identical manner in all centers. Under sterile conditions the skin was anesthetized with 1% lidocaine without epinephrine. A 15-gauge needle was used to aspirate and multiply puncture the walls of the cyst by means of digital palpation to direct the needle and decompress the mass. Palmar carpal and palmar digital ganglions were punctured; care was taken to direct the needle away from the radial artery and digital neurovascular bundle respectively.

Forty-one patients had their wrists immobilized after treatment. By prior agreement all 29 patients from San Diego and one half of the patients from Madison had 3 weeks of immobilization after aspiration-puncture. All 34 patients from Stockton and one half

From the University of California, San Diego Medical Center, San Diego, Calif. and the University of Wisconsin Medical Center, Madison, Wis.

Received for publication Sept. 15, 1986; accepted in revised form Nov. 22, 1986.

No benefits in any form have been received or will be received from a commercial party related directly or indirectly to the subject of this article.

Reprint requests: Richard H. Gelberman, M.D., Division of Orthopaedic Surgery, Massachusetts General Hospital, ACC 427, Boston, MA 02114.

Table I

	Patient	Ganglions	Mean age	M	F	Mean duration (mo)	Mean follow-up (mo)	Mean time to recurrence (mo)	Success rate (%)
Total	85	87	33	51	34	14.9 (1-120)	22 (6-46)	2.8 (1-7)	36
Dorsal carpal	58	60	30 (6-79)	39	19	19 (1-120)	22 (6-36)	3 (1-7)	27
Palmar carpal	14	14	39 (11-63)	7	7	3 (1-6)	28 (18-46)	3.5 (1-4)	43
Palmar digital	12	13	44 (25-58)	5	7	7.5 (1-24)	17 (9-32)	1.5 (1-2)	69

Table II. Success of cyst puncture with and without immobilization (%)

	Early mobilization	Three weeks' immobilization
Total	29 (13/45)	43 (18/42)
Dorsal carpal	13 (4/30)	40 (12/30)*
Palmar carpal	38 (3/8)	50 (3/6)
Palmar digital	86 (6/7)	50 (3/6)

*0.02 ≤ *p* ≤ 0.05.

of the patients from Madison were placed in the early mobilization group. Patients with carpal ganglions had splints applied for 3 weeks in a plaster cast or thermoplast splint, with the wrist in 10° of extension. Patients with digital ganglions had their wrists in splints for 3 weeks in a plaster cast with an aluminum outrigger, with the metacarpophalangeal (MP) joints in 20° of flexion and the interphalangeal (IP) joints free. Forty-four patients had early mobilization after aspiration-puncture. Patients in this group were not treated by any immobilization and were allowed to resume full preaspiration activities.

Results

There were no differences in age, sex, follow-up, or time to recurrence between dorsal carpal, palmar carpal, and palmar digital ganglions (Table I). The success rate for all ganglions was 36% (31/87). Twenty-seven percent (16/60) of dorsal carpal ganglions, 43% (6/14) of palmar carpal ganglions, and 69% (9/13) of palmar digital ganglions were successfully treated by aspiration-puncture.

Dorsal carpal ganglions of less than or equal to 6 months duration were successfully treated by aspiration-puncture in 41% (11/27) of cases compared with 16% (4/25) for dorsal carpal ganglions of more than 6 months duration. This trend, however, did not

reach statistical significance ($0.05 > p < 0.1$). There was no difference in duration of symptoms between successfully treated ganglions and recurrences.

Immobilization significantly affected outcome as can be seen in Table II. There were 30 ganglions in both immobilization and early mobilization groups. Forty percent (12/30) of dorsal carpal ganglions immobilized for 3 weeks had a successful outcome compared with 13% (4/30) of dorsal carpal ganglions in the early mobilization group ($p < 0.05$). Seventy-five percent of successfully treated dorsal carpal ganglions were immobilized compared with 40% of recurrent ganglions ($p < 0.05$).

Discussion

An overall success rate of 36% in 87 ganglions treated by aspiration, multiple puncture, and digital compression is lower than most previous reports of similar conservative methods. In 1928 Carp and Stout reported a 75% (6/8) success rate in ganglions treated by aspiration and injection of iodine or carbolic acid. In the same series 78% (7/9) of ganglions manually ruptured and 58% (7/12) of ganglions followed-up for 3 years without active treatment had no recurrence of the mass. These numbers are small, and there was no attempt to differentiate between dorsal carpal, palmar carpal, and digital ganglions. In 1954 McEvedy⁴ reported a 41% spontaneous resolution rate in 21 ganglions followed-up for 10 years without active treatment. In addition, he reported an 82% success rate in 43 ganglions treated with the injection of a sclerosing agent (either 10% morrhuate or ethanolamine). No distinction was made between different types of ganglions, and there have been no similar studies reporting equivalent success rates. Nelson, Sawmiller, and Phalen⁶ in 1972, reported on the conservative treatment of 109 patients. They noted a 66% success rate in 44 patients

treated with digital pressure and rupture and a 40% success rate in 146 patients treated with injection of various combinations of steroids. After their report we began offering aspiration-puncture and digital rupture as definitive treatment of patients with carpal and digital ganglions who wished to avoid or delay surgery. The purpose of this report was to examine our experience with this single treatment method.

In this study 27% of all dorsal carpal ganglions treated by aspiration-puncture had a successful outcome. The addition of 3 weeks of immobilization to our treatment protocol increased our success rate to 41% ($p < 0.05$). Aspiration puncture followed by immobilization may promote sufficient scar formation to obliterate the one-way valve mechanism postulated by Andren and Eiken.⁷ We were unable to identify those characteristics in individual patients that served as accurate prognostic indicators for successful conservative management. Dorsal carpal ganglions of less than or equal to 6 months duration responded more favorably to aspiration-puncture than did ganglions of more than 6 months duration. This trend could be explained, however, by a difference in immobilization between the two groups. In the shorter duration group 81% of successfully treated dorsal carpal ganglions and 35% of the unsuccessfully treated carpal ganglions were immobilized ($p < 0.05$). In the more than 6 months duration group no such difference existed. Our 69% (9/13) success rate with palmar digital ganglions is consistent with Bruner's² report of 15 ganglions treated with aspiration-puncture without immobilization. He reported only one recurrence. There are no studies documenting a communication between palmar digital ganglions and the flexor tendon synovial sheath. The absence of such a communication could explain the lower recurrence rate seen in this subgroup of ganglions.

Safety and cost-effectiveness make aspiration-puncture an attractive alternative to operative treatment. There were no complications in this series. Operative treatment of carpal ganglions is reported to be successful in 95% of cases if excision of the cyst and a cuff of capsular tissue is performed with the patient under adequate anesthesia and tourniquet control.^{6,8} On the basis of our results we believe that patients with carpal ganglions who wish to avoid or delay operative treatment should be advised that aspiration-puncture followed by 3 weeks of immobilization will be successful in approximately 50% of cases (23/50). Patients with palmar digital ganglions can be even more optimistic about their chances for a successful outcome with conservative management.

REFERENCES

1. Webber B. The simple ganglion. (Letter) *Lancet* 1954; 266:265.
2. Bruner J. Treatment of "sesamoid" synovial ganglia of the hand by needle rupture. *J Bone Joint Surg [Am]* 1963;45:1689-90.
3. Holm P, Pandey S. Treatment of ganglia of the hand and wrist with aspiration and injection of hydrocortisone. *Hand* 1973;5:63-7.
4. McEvedy B. The simple ganglion. *Lancet* 1954;266:135.
5. Carp L, Stout A. A study of ganglion. *Surg Gynecol Obstet* 1928;47:460-8.
6. Nelson C, Sawmiller S, Phalen G. Ganglions of the hand and wrist. *J Bone Joint Surg [Am]* 1972;54:1459-64.
7. Andren L, Eiken B. Arthrographic studies of wrist ganglions. *J Bone Joint Surg [Am]* 1971;53:299-302.
8. Angelides A, Wallace P. The dorsal ganglion of the wrist: Its pathogenesis, gross and microscopic anatomy, and surgical treatment. *J HAND SURG* 1976;1:228-35.